

Module 2: Fundamentals of Cyber Security

1. Introduction

Cyber security, also known as computer security or digital security, focuses on protecting computer systems, networks, programs, and data from unauthorized access, damage, theft, or disruption. With the growth of the internet, smart devices, and interconnected systems, cyber security has become critical for individuals, businesses, and governments to maintain privacy, data integrity, and service availability.

2. Core Concepts and Objectives of Cyber Security

The foundation of cyber security rests on the **CIA Triad**:

- **Confidentiality:** Ensuring that information is accessible only to authorized users. This protects sensitive data from unauthorized disclosure. Tools like encryption and access controls are primary methods to guarantee confidentiality.
- **Integrity:** Maintaining accuracy and consistency of data by preventing unauthorized modification or tampering. This includes measures for data validation and detection of unauthorized changes.
- **Availability:** Guaranteeing that systems, applications, and data are accessible and operational when needed by authorized users. Protection against attacks like Denial of Service (DoS) is essential to ensure availability.

3. Key Areas of Cyber Security

- **Network Security:** Protecting data and resources as they travel across or are accessed via networks using firewalls, intrusion detection systems, and encryption.

- **Application Security:** Ensuring software and applications are safeguarded against threats such as viruses, malware, and hacking attempts by implementing secure coding practices and ongoing vulnerability testing.
- **Information Security:** Protecting the confidentiality, integrity, and availability of data in any form.
- **Operational Security:** Managing and protecting digital processes and permissions, including access controls and incident response.
- **End-user Education:** Training users to recognize threats like phishing, social engineering, and safe practices to minimize risk.
- **Physical Security:** Protecting hardware and physical devices from tampering or theft, complementing digital security measures.

4. Common Cyber Security Threats

- **Malware:** Malicious software such as viruses, worms, ransomware, and spyware designed to harm or exploit systems.
- **Phishing:** Fraudulent attempts to obtain sensitive information by masquerading as trustworthy entities.
- **Denial of Service (DoS) and Distributed DoS (DDoS):** Attacks that overwhelm systems, rendering them unavailable to legitimate users.
- **Man-in-the-Middle (MitM) Attacks:** Intercepting communication between two parties to steal or alter data.
- **Zero-Day Exploits:** Attacks that exploit unknown or unpatched vulnerabilities.
- **Social Engineering:** Manipulating individuals to divulge confidential information.

5. Cyber Security Best Practices

- **Authentication and Authorization:** Strong, multi-factor authentication (MFA) to verify user identities, and strict access control policies.
- **Encryption:** Securing data at rest and in transit to prevent unauthorized access.
- **Regular Patch Management:** Timely updating of systems and applications to fix vulnerabilities.
- **Security Monitoring and Incident Response:** Detecting, responding to, and recovering from security incidents swiftly.
- **Backup and Recovery:** Regular data backups to restore systems in case of data loss or ransomware attacks.
- **Security by Design:** Incorporating security measures from the initial stages of system and software development to reduce vulnerabilities.

6. Summary

Cyber security is a broad and evolving field aimed at protecting digital assets from a wide array of threats. Mastery of its fundamentals—including the CIA Triad, types of threats, and best practices—is essential for both IT professionals and users to safeguard systems and maintain trust in digital technologies.